extend the original image tokens by concatenating the visually cropped image tokens, as illustrated in Fig. 4. We use this concatenation approach when applying all our methods to MLLMs.

Relative Attention ViCrop (`rel-att`). In this method, we directly compute the relative attention $A_{rel}(x, q)$ defined in Sec. 4 for each image-question pair (x, q). We then select a target layer (in LLM and connector) based on a small held-out set of samples in TextVQA and use its relative attention as the importance map for visual cropping (discussed below). We ablate on the choice of layer in Sec. 6.

Gradient-Weighted Attention ViCrop (`grad-att`). The relative attention runs an additional generic instruction through the MLLM to normalize the answer-to-image attention and emphasize semantically relevant attention. As an alternative that does not require a second forward pass, we consider using the gradients to normalize attention, because the gradient of the model's decision with respect to an attention score shows how sensitive the decision is to changes in that attention, hence how semantically relevant the attention is for answering the question. To get a differentiable representation of the model's decision, we consider the logarithm of the maximum output probability at the starting answer token, $v = \log \text{softmax}(z(x, q))_{t^*} \in \mathbb{R}$, where $z \in \mathbb{R}^D$ is the output logit of the LLM at the starting answer position, D the vocabulary size, and $t^* = \arg\max_t z_t$. Then, following our notation in Sec. 4, we can compute the gradient-weighted versions of answer-to-token attention $\tilde{A}_{st}(x, q) = A_{st}(x, q) \odot \sigma(\nabla_{A_{st}} v(x, q))$ and token-to-image attention $\tilde{A}_{ti}(x, q) = A_{ti}(x) \odot \sigma(\nabla_{A_{ti}} v(x, q))$, where $\odot$ is element-wise product and $\sigma(w) = \max(0, w)$. We remove negative gradients because they correspond to tokens that if attended to will reduce the model certainty, hence often distracting locations Selvaraju et al. (2017). Finally, we compute the gradient-weighted answer-to-image attention as the following tensor product $\tilde{A}_{si}(x, q) = \tilde{A}_{st}(x, q) \otimes \tilde{A}_{ti}(x, q) \in \mathbb{R}^{L \times L_c \times 1 \times N^2}$. We will select the same target layer identified in `rel-att` from $\tilde{A}_{si}(x, q)$ as the importance map for visual cropping.

Input Gradient ViCrop (`pure-grad`). In this method, we seek to find the relevant regions on the image directly using the gradient of the MLLM's decision with respect to the input image. Compared to the previous attention-based methods, `pure-grad` is more versatile because it does not rely on the Transformer-based architecture. Specifically, for each image-question pair (x, q), we will compute $G(x, q) = \|\nabla_x v(x, q)\|_2$, where $v(x, q)$ is the logarithm of the maximum output probability of the LLM at the starting answer token (as defined in `grad-att` above), and the L2-norm is taken over the image channel dimension. However, gradients sometimes show high values in entirely constant-color regions (*e.g.*, blue skies). Given that these non-edge regions do not contain any visual details, we will explicitly diminish them in G. To that end, we first apply a 3×3-size Gaussian high-pass filter to the image, followed by a median filter of the same size to reduce salt-and-pepper noise. The resulting filtered image is then thresholded at its spatial median value to become a binary mask and is element-wise multiplied by G. Finally, the edge-emphasized G is spatially average-pooled into the $N \times N$ patches of the MLLM to become an importance map for visual cropping.

Bounding Box Selection for Visual Cropping. To convert the importance map (from each of the methods described above) to a bounding box, we use sliding windows of different sizes inspired by object detection literature Redmon et al. (2016). Specifically, for each MLLM, we define a set of windows with sizes equal to a multiple of the input image resolution of the MLLM. The multiples are in $\{1, 1.2, \ldots 2\}$. We slide each window over the image with a stride of 1 and find its best position where the sum of the importance map inside the window is maximized. After selecting the best position per window, we choose the window whose internal sum has the largest difference from the average internal sum of its adjacent positions. This latter step is a heuristic to avoid choosing too small or too large windows (notice that in both cases, moving the window slightly left/right or up/down will not change its internal sum significantly). The chosen window is then cropped from the image, resized to the input image resolution of the MLLM, and provided to the MLLM in addition to the image-question pair.

High-Resolution Visual Cropping. In one of the benchmarks we consider in this work, V* Wu and Xie (2023), the images are of very high resolution (always more than 1K) and consequently, the resized input image provided to the MLLM might completely lose the visual concept of interest for a question. To mitigate this, on this benchmark, we employ a two-stage strategy. In the first stage, we divide images into non-overlapping blocks of smaller than 1024×1024 with an aspect ratio close to 1, compute the importance map separately for each block according to the ViCrop methods, and then re-attach the blocks back together. In the second stage, we find the bounding box for visual

Figure 5: Examples of `rel-att` helping MLLMs correct their mistakes (cyan-colored bounding box shows cropped region by `rel-att`; zoom-in insets are displayed for better readability).

cropping on this re-attached importance map exactly as described before and provide the original image-question pair with the resized cropped image to the MLLM.

6 ViCrop Method Analysis

In this section, we apply our proposed visual cropping methods to two open-source SOTA MLLMs, InstructBLIP (Vicuna-7B) (Dai et al., 2023) and LLaVA-1.5 (Vicuna-7B) (Liu et al., 2023a). We evaluate their effectiveness in improving the perception of smaller visual concepts on 4 detail-sensitive datasets (TextVQA [2] (Singh et al., 2019), V* (Wu and Xie, 2023), POPE (Li et al., 2023c), DocVQA (Mathew et al., 2021)), and their ability to maintain performance on larger visual concepts in 3 general-purpose datasets containing mostly large objects (GQA (Hudson and Manning, 2019), AOKVQA (Schwenk et al., 2022), VQAv2 (Goyal et al., 2017)). InstructBLIP uses the hyper-parameters $N = 16, m = 15, k = 2$ and input image resolution of 224×224. LLaVA-1.5 uses $N = 24, m = 14$ and input image resolution of 336×336. When reporting accuracy, we compute VQA-score[3] for all benchmarks except GQA. For GQA, we compute accuracy using the official code.[4]. See Appendices A to C for further details about implementation, datasets, and prompts.

ViCrop Improves VQA Accuracy. In Fig. 5, we show a few examples of the ViCrop helping the MLLM correct itself (more examples in Appendix G), and in Tab. 2, we report the accuracy of the proposed ViCrop methods on the VQA benchmarks. We observe that all methods significantly improve the accuracy of the original MLLMs (*no cropping*) on detail-sensitive benchmarks, without requiring any training, while maintaining the MLLMs' performance on benchmarks with larger visual concepts. Thus, the accuracy gain on fine details (most notably in TextVQA and V*) does not seem to come at the cost of accuracy on larger visual details and relations. We also observe that the accuracy gain for LLaVA-1.5 is more substantial than for InstructBLIP. This can be explained by the

[2]† In TextVQA evaluation, we do not provide externally extracted OCR tokens to the MLLM since we want to measure its true perception, this differs from the setup in the original paper. See more discussion in Appendix A.

[3] https://visualqa.org/evaluation.html

[4] https://cs.stanford.edu/people/dorarad/gqa/evaluate.html

Table 2: Accuracy of the proposed ViCrop methods on visual question answering benchmarks.

Model		Smaller Visual Concepts				Larger Visual Concepts		
		TextVQA†	V*	POPE	DocVQA	AOKVQA	GQA	VQAv2
LLaVA-1.5	no cropping	47.80	42.41	85.27	15.97	59.01	60.48	75.57
	rel-att	55.17	**62.30**	**87.25**	19.63	**60.66**	60.97	**76.51**
	grad-att	**56.06**	57.07	87.03	**19.84**	59.94	**60.98**	76.06
	pure-grad	51.67	46.07	86.06	17.70	59.92	60.54	75.94
InstructBLIP	no cropping	33.48	35.60	84.89	9.20	60.06	49.41	76.25
	rel-att	45.44	**42.41**	86.64	9.95	61.28	49.75	**76.84**
	grad-att	**45.71**	37.70	**86.99**	**10.81**	**61.77**	**50.33**	76.08
	pure-grad	42.23	37.17	86.84	8.99	61.60	50.08	76.71

Table 3: Ablation study on the choice of layer and the use of high-resolution visual cropping.

Model		Choice of Layer			High-Resolution ViCrop		
		Selective	Average	Δ	w/ High-Res	w/o High-Res	Δ
LLaVA-1.5	no cropping	47.80	–	–	42.41	42.41	–
	rel-att	55.17	55.45	+0.28	62.30	47.64	-14.66
	grad-att	56.06	56.26	+0.20	57.07	49.74	-7.33
	pure-grad	51.67	–	–	46.07	45.03	-1.04
InstructBLIP	no cropping	33.48	–	–	35.60	35.60	–
	rel-att	45.44	44.40	-1.04	42.41	38.74	-3.67
	grad-att	45.71	44.98	-0.73	37.70	42.41	+4.71
	pure-grad	42.23	–	–	37.17	42.41	+5.24

fact that InstructBLIP only trains its connector and not its backbone LLM during tuning—the LLM does not adapt to use the image tokens, rather the image tokens are adapted to optimally prompt the LLM—and therefore the LLM cannot effectively use the additional (cropped) image tokens provided through visual cropping. Nonetheless, the results show that ViCrop can be effectively applied to different MLLMs, and is a promising inference-time solution for mitigating the perception limitation observed in Sec. 3.

Ablation Study on the Choice of Layer. To understand the importance of the choice of an informative layer for rel-att and grad-att (as discussed in Sec. 5), in Tab. 3 we compare the accuracy of these methods when simply taking the average of all layers in A_{rel} and $\tilde{A}_{si}$, respectively, on TextVQA. We observe that rel-att is robust to this choice and grad-att declines about 3.5 percentage points in accuracy. Importantly, both methods still improve the MLLMs' accuracy even when using the layer average, suggesting that averaging is a suitable choice in the absence of any data for selecting a layer.

Ablation Study on the High-Resolution ViCrop. In Sec. 5, we proposed a two-stage strategy for processing the very high-resolution images in the V* benchmark. To see how effective this strategy is, in Tab. 3 we compare the accuracy of ViCrop methods with and without this high-resolution strategy on V*. We observe that while this strategy is very beneficial to LLaVA-1.5, it declines the performance of grad-att and pure-grad for InstructBLIP. However, all methods, with and without this strategy, still improve the MLLMs' accuracy.

ViCrop with External Tools. In addition to the internal ViCrop methods, we also considered the use of external off-the-shelf models to find the region of interest in an image for visual cropping. Specifically, we utilized SAM (Kirillov et al., 2023), YOLO (Redmon et al., 2016), and CLIP (Radford et al., 2021) to find the most relevant part of an image to a given question (details of these external ViCrop methods are provided in Appendix F). In Tab. 4, we compare the accuracy of external ViCrop methods to the internal methods on TextVQA. While external models are also effective in improving the accuracy of MLLMs, they are weaker than all the proposed internal ViCrop methods, thus we did not explore them further.